COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	10/5/2026	**DATE SUBMITTED:**	10/5/2026
NAME:	Momo, Borge Rudolf A.	**INSTRUCTOR:**	Engr. Quejado

ACTIVITY NO. 10.1 **Install, Configure and Manage Enterprise Log Monitoring via Ansible**

1. PROCEDURE & OUTPUT

1. Create a playbook that:
a. Install and configure Elastic Stack in separate hosts (Elastic Search, Kibana, Logstash)
2. Apply the concept of creating roles.
3. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)

The Tree to verify that concept of creating roles was applied.

```
momo1@workstation:~/CPE212_MOMO/activity10$ tree
.
└── roles
    ├── elasticsearch
    │   └── tasks
    ├── kibana
    │   └── tasks
    └── logstash
        └── tasks

8 directories, 0 files
momo1@workstation:~/CPE212_MOMO/activity10$
```

```
momo1@workstation:~/CPE212_MOMO/activity10$ cat inventory.ini
[elasticsearch]
192.168.56.102 ansible_user=bmomo

[kibana]
192.168.56.103 ansible_user=momo2

[logstash]
192.168.56.104 ansible_user=momo3
```

This is the inventory.ini, each managed node has their own toolset.

INSTALLING ELASTICSEARCH

```
---
- name: Install the packages needed for Elasticsearch
  ansible.builtin.apt:
    name:
      - apt-transport-https
      - wget
      - gnupg
    state: present
    update_cache: yes
```

This part just installs the basic packages needed to download and install Elasticsearch on the server.

```
- name: Add the Elasticsearch signing key
  ansible.builtin.shell: |
    wget -qO - https://artifacts.elastic.co/GPG-KEY-elasticsearch | gpg --dearmor -o /usr/share/keyrings/elasticsearch-keyring.gpg
  args:
    creates: /usr/share/keyrings/elasticsearch-keyring.gpg
```

This part adds the Elasticsearch security key so the system can verify the packages from Elastic.

```
- name: Add the Elasticsearch package repository
  ansible.builtin.apt_repository:
    repo: "deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.gpg] https://artifacts.elastic.co/packages/9.x/apt stable main"
    state: present
    filename: elastic
```

This part adds the official Elasticsearch repository so the server can get the Elasticsearch package.

```
- name: Install Elasticsearch on the server
  ansible.builtin.apt:
    name: elasticsearch
    state: present
    update_cache: yes
```

This part installs Elasticsearch on the server using the repository that was added earlier.

```
- name: Start Elasticsearch and enable it at boot
  ansible.builtin.service:
    name: elasticsearch
    state: started
    enabled: yes
```

This part starts the Elasticsearch service and makes sure it will automatically start when the server boots.

INSTALLING KIBANA

```
---
- name: Install the packages needed for Kibana
  ansible.builtin.apt:
    name:
      - apt-transport-https
      - wget
      - gnupg
    state: present
    update_cache: yes
```

This part installs the basic packages needed to download and install Kibana.

```
- name: Add the Kibana signing key
  ansible.builtin.shell: |
    wget -qO - https://artifacts.elastic.co/GPG-KEY-elasticsearch | gpg --dearmor -o /usr/share/keyrings/kibana-keyring.gpg
  args:
    creates: /usr/share/keyrings/kibana-keyring.gpg
```

This part adds the security key so the system can verify the Kibana package.

```
- name: Add the Kibana package repository
  ansible.builtin.apt_repository:
    repo: "deb [signed-by=/usr/share/keyrings/kibana-keyring.gpg] https://artifacts.elastic.co/packages/9.x/apt stable main"
    state: present
    filename: kibana
```

This part adds the official Elastic repository where the Kibana package can be downloaded.

```
- name: Install Kibana on the server
  ansible.builtin.apt:
    name: kibana
    state: present
    update_cache: yes
```

This part installs Kibana on the Ubuntu server.

```
- name: Start Kibana and enable it at boot
  ansible.builtin.service:
    name: kibana
    state: started
    enabled: yes
```

This part starts Kibana and makes it automatically start when the server boots.

INSTALLING LOGSTASH

```
---
- name: Install the packages needed for Logstash
  ansible.builtin.apt:
    name:
      - apt-transport-https
      - wget
      - gnupg
    state: present
    update_cache: yes
```

This part installs the basic packages needed to download and install Logstash.

```
- name: Add the Logstash signing key
  ansible.builtin.shell: |
    wget -qO - https://artifacts.elastic.co/GPG-KEY-elasticsearch | gpg --dearmor -o /usr/share/keyrings/logstash-keyring.gpg
  args:
    creates: /usr/share/keyrings/logstash-keyring.gpg
```

This part adds the security key so the system can verify the Logstash package.

```
- name: Add the Logstash package repository
  ansible.builtin.apt_repository:
    repo: "deb [signed-by=/usr/share/keyrings/logstash-keyring.gpg] https://artifacts.elastic.co/packages/9.x/apt stable main"
    state: present
    filename: logstash
```

This part adds the official Elastic repository where the Logstash package can be downloaded.

```
- name: Install Logstash on the server
  ansible.builtin.apt:
    name: logstash
    state: present
    update_cache: yes
```

This part installs Logstash on the Ubuntu server.

```
- name: Start Logstash and enable it at boot
  ansible.builtin.service:
    name: logstash
    state: started
    enabled: yes
```

This part starts Logstash and makes it automatically start when the server boots.

SITE.YML

```
---
- name: Install Elasticsearch
  hosts: elasticsearch
  become: yes
  roles:
    - elasticsearch
```

This part tells Ansible to use the Elasticsearch server and run the Elasticsearch role on it.

```
- name: Install Kibana
  hosts: kibana
  become: yes
  roles:
    - kibana
```

This part tells Ansible to use the Kibana server and run the Kibana role on it.

```
- name: Install Logstash
  hosts: logstash
  become: yes
  roles:
    - logstash
```

This part tells Ansible to use the Logstash server and run the Logstash role on it.

4. Show an output of the installed Elastic Stack for both Ubuntu and CentOS.

ElasticSearch

```
PLAY [Install Elasticsearch] ***************************************************

TASK [Gathering Facts] *********************************************************
[WARNING]: Host '192.168.56.102' is using the discovered Python interpreter at '/usr/bin/p
 cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core
tion.
ok: [192.168.56.102]

TASK [elasticsearch : Install the packages needed for Elasticsearch] ***********
ok: [192.168.56.102]

TASK [elasticsearch : Add the Elasticsearch signing key] ***********************
changed: [192.168.56.102]

TASK [elasticsearch : Add the Elasticsearch package repository] ****************
changed: [192.168.56.102]

TASK [elasticsearch : Install Elasticsearch on the server] *********************
changed: [192.168.56.102]

TASK [elasticsearch : Start Elasticsearch and enable it at boot] ***************
changed: [192.168.56.102]
```

Kibana

```
PLAY [Install Kibana] *********************************************************

TASK [Gathering Facts] *********************************************************
[WARNING]: Host '192.168.56.103' is using the discovered Python interpreter at '/usr/bin
d cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-c
ation.
ok: [192.168.56.103]

TASK [kibana : Install the packages needed for Kibana] *****************************
changed: [192.168.56.103]

TASK [kibana : Add the Kibana signing key] *************************************
changed: [192.168.56.103]

TASK [kibana : Add the Kibana package repository] *********************************
changed: [192.168.56.103]

TASK [kibana : Install Kibana on the server] ***********************************
changed: [192.168.56.103]

TASK [kibana : Start Kibana and enable it at boot] *********************************
changed: [192.168.56.103]
```

Logstash

```
PLAY [Install Logstash] ********************************************************

TASK [Gathering Facts] *********************************************************
[WARNING]: Host '192.168.56.104' is using the discovered Python interpreter at '/usr/bi
d cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
ation.
ok: [192.168.56.104]

TASK [logstash : Install the packages needed for Logstash] *************************
changed: [192.168.56.104]

TASK [logstash : Add the Logstash signing key] *********************************
changed: [192.168.56.104]

TASK [logstash : Add the Logstash package repository] *****************************
changed: [192.168.56.104]

TASK [logstash : Install Logstash on the server] *******************************
changed: [192.168.56.104]

TASK [logstash : Start Logstash and enable it at boot] *****************************
changed: [192.168.56.104]
```

PLAY RECAP

```
PLAY RECAP *********************************************************************
192.168.56.102             : ok=6    changed=4    unreachable=0    failed=0
192.168.56.103             : ok=6    changed=5    unreachable=0    failed=0
192.168.56.104             : ok=6    changed=5    unreachable=0    failed=0

momo1@workstation:~/CPE212_MOMO/activity10$
```

Verification that it was installed on CentOS as well as Ubuntu:

CentOS (ElasticSearch)

```
momo1@workstation:~/CPE212_MOMO/activity10$ ssh bmomo@192.168.56.102
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Mon Oct  5 19:42:46 2026 from 192.168.56.101
[bmomo@localhost ~]$ sudo systemctl status elasticsearch
[sudo] password for bmomo:
● elasticsearch.service - Elasticsearch
     Loaded: loaded (/usr/lib/systemd/system/elasticsearch.service; enabled;
     Active: active (running) since Mon 2026-10-05 19:29:05 PST; 14min ago
       Docs: https://www.elastic.co
   Main PID: 6423 (server-launcher)
      Tasks: 105 (limit: 22965)
     Memory: 2.0G (peak: 2.1G)
        CPU: 2min 12.915s
     CGroup: /system.slice/elasticsearch.service
             ├─6423 /usr/share/elasticsearch/lib/tools/server-launcher/serve
             ├─6503 /usr/share/elasticsearch/jdk/bin/java -Des.networkaddres
             └─6528 /usr/share/elasticsearch/modules/x-pack-ml/platform/linu
```

Ubuntu Server1 (Kibana)

```
momo1@workstation:~/CPE212_MOMO/activity10$ ssh momo2@192.168.56.103
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

149 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

*** System restart required ***
Last login: Mon Oct  5 11:35:19 2026 from 192.168.56.101
momo2@Server1:~$ sudo systemctl status kibana
● kibana.service - Kibana
     Loaded: loaded (/usr/lib/systemd/system/kibana.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-10-05 11:35:21 UTC; 9min ago
 Invocation: 5bd3bc7ef815488dbd3d55e34e4f2bd9
       Docs: https://www.elastic.co
   Main PID: 33096 (MainThread)
      Tasks: 11 (limit: 4024)
     Memory: 516.4M (peak: 791.2M)
        CPU: 34.562s
     CGroup: /system.slice/kibana.service
             └─33096 /usr/share/kibana/bin/../node/default/bin/node /usr/share/kibana>
```

Ubuntu Server2 (Logstash)

```
momo3@Server2:~$ sudo systemctl status logstash
● logstash.service - logstash
     Loaded: loaded (/usr/lib/systemd/system/logstash.service; enabled; preset: enabl
     Active: active (running) since Mon 2026-10-05 11:47:52 UTC; 10s ago
 Invocation: 36e2758a6f7c456fa30a2bd76edfe598
   Main PID: 9996 (java)
      Tasks: 21 (limit: 1862)
     Memory: 253.2M (peak: 253.2M)
        CPU: 18.555s
     CGroup: /system.slice/logstash.service
             └─9996 /usr/share/logstash/jdk/bin/java -Xms1g -Xmx1g -Djava.awt.headles
```

Analysis:
We can see that they all worked, I used the systemctl status command to verify that all three were installed and working properly.

5. Make sure to create a new repository in GitHub for this activity.



https://github.com/BorgeMomo/CPE212_MOMO

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:
1. What are the benefits of having a log monitoring tool?

- A log monitoring tool helps us track what is happening in our servers and systems and it makes it easier to find errors and unusual activities before they become bigger issues. It also helps administrators troubleshoot problems faster and keep the system running properly.

CONCLUSION

In conclusion, I learned how to install and manage Elastic Stack using Ansible and I used separate servers for Elasticsearch, Kibana, and Logstash, and created roles to organize my playbook. I also verified that all three services were installed and running properly on their assigned servers. Overall, I learned that Ansible made the installation easier and helped me manage the log monitoring tools in a more organized way.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Google AI**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Understanding of Concepts**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **The tool was only used to understand some concepts but the answers and procedures were answered on my own.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.